Rana Mozumder

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Education

Vanderbilt University (VU) *Nashville, TN, USA*

Ph.D. in Biomedical Engineering *since Aug 2022*

- **Research:** Cognitive neuroscience, Neuroengineering, NeuroAI.
- **Key Courses:** Special topics in Deep Learning, Advanced Quantitative Image Analysis, Analysis of fMRI, Quantitative Methods in BME, Fundamentals of Neuroscience II.

Bangladesh University of Engineering and Technology (BUET) *Dhaka, Bangladesh*

Bachelor of Science, Biomedical Engineering *2016-2021*

- **CGPA:** 3.78/4.00
- **Research:** Biomedical signal processing.
- **Key Courses:** Digital Signal Processing, Random Signal Processing, Medical Imaging, Biomedical Instrumentation, Tissue Engineering.

Research Interests

I aim to understand the fundamental principles of brain function and apply this knowledge to develop novel therapeutic strategies for neurodegenerative diseases and practical brain-computer interfaces. My interests lie at the intersection of cognitive neuroscience, brain-inspired computation, and biomedical imaging.
Keywords: *Neuroscience, Biomedical Engineering, NeuroAI, Biomedical Imaging, Electrophysiology, Dynamical systems.*

Research Experiences

Department of Biomedical Engineering, VU *Nashville, TN, USA*

Graduate Research Assistant - Constantinidis Lab *Aug 2022- Present*

- **Core Responsibilties**
  - Leading optogenetic experiments on non-human primates (NHPs) in the lab.
  - Designing new experiments, training NHPs on novel tasks.
  - Conducting multimodal recording: electrophysiological recording with various laminar probes, e.g., Plexon, Diagnostic Bio Chips, Neuropixels probes, and behavioral data (eye-tracking).
  - Preprocessing and data analysis.
  - Analyzing CT and MR images (Registration, Segmentation) for surgery.
- **Selected Projects**
  - **Optotagging Somatostating interneurons in macaque prefrontal cortex.** Developing and applying optogenetic approaches to identify (optotag) and manipulate somatostatin (SST) interneurons in the dorsolateral prefrontal cortex of behaving macaques. By combining cell-type-specific optical stimulation with electrophysiological recordings, this project investigates how inhibitory interneurons regulate local circuit dynamics underlying working memory. This work enables causal, cell-type-specific interrogation of cortical microcircuits for the first time in a non-human primate model.
  - **Asynchronous firing and off states in working memory maintenance.** Examined whether persistent activity observed during working memory is an artifact of trial-averaging or reflects true neural dynamics. Using single-trial analyses (raster plots, inter-spike interval distributions) alongside population decoding and machine learning classifiers, we demonstrate that working memory can be supported by asynchronous and sparse firing patterns with intermittent “off” states. Read the paper *here*.
  - **Repeatable, low-drift recordings in behaving non-human primates using flexible microelectrodes.** Collaborating with the *Gonzales Lab* to develop next-generation flexible microelectrode arrays for stable, long-term neural recordings in non-human primates. This work focuses on minimizing tissue damage and electrode drift while maintaining high signal quality during behavior. Current efforts involve acute validation, with the long-term goal of enabling chronic, large-scale recordings for studying neural dynamics across extended timescales. Read the paper *here*.
  - **Integration of audiovisual motion in dorsolateral prefrontal cortical neurons.** Demonstrated that neurons in the dorsolateral prefrontal cortex integrate auditory and visual motion signals, revealing that multisensory processing extends beyond classical sensory areas into higher-order cognitive regions. This work highlights the role of prefrontal circuits in combining sensory evidence to support flexible decision-making and working memory. Read the paper *here*.
  - **Contributions of narrow- and broad-spiking prefrontal and parietal neurons on working memory tasks.** Classified neurons into putative inhibitory (narrow-spiking) and excitatory (broad-spiking) populations based on waveform features, and quantified their differential contributions to working memory tasks. Results reveal distinct functional roles of these cell types in encoding and maintaining task-relevant information. Read the paper *here*
  - **Single-neuron and population measures of neuronal activity in working memory tasks.** Compared traditional single-neuron analyses with population-level decoding approaches to assess how information is represented in prefrontal cortex. This work demonstrates that population-level methods capture distributed coding and latent structure not apparent at the single-neuron level, emphasizing the importance of high-dimensional neural representations. Read the paper *here*

Department of Biomedical Engineering, BUET *Dhaka, Bangladesh*

Undergraduate Research Student *2018 - 2021*

- **Undergrad Thesis**
  - **Subject Independent Mental Task Classification using Energy based Features from EEG Signal.** In this work two methods were investigated for classification. Firstly, the signal was divided into five different frequency bands and Entropy and Log-variance of signals were calculated to construct the feature vector. And secondly, Variational Mode Decomposition (VMD) analysis was performed and then, same energy related features were extracted from the Intrinsic Mode Functions found after VMD analysis to form the feature vector. In both cases k-Nearest Neighbors was used as the classifier. Thesis report

Teaching and Internship Experiences

Vanderbilt Summer Academy for Talented Youth *Nashville, TN, USA*

Instructor for *The Thinking Brain: Unlocking Memory, Decisions, and the Prefrontal Cortex* *June 2026*

- Designed and taught a neuroscience course covering brain function, working memory, decision-making, aging, and neurotechnologies.
- Developed hands-on MATLAB activities for neural data analysis and visualization.
- Led inquiry-based activities, including working memory games and scientific paper discussions.
- Mentored student teams on neuroscience projects involving experimental design, brain–computer interfaces, and optogenetics.

Biomedical Engineering Department, VU *Nashville, TN, USA*

Graduate Teaching Assistant for *Biomedical Instrumentation II* course *Fall'23, Fall'24*

- Guided senior-level students in building robust hardware systems for recording biophysical signals such as ECG.
- Assisted in designing and debugging microcontroller codes to collect, preprocess, and visualize physiological data.
- Led hands-on lab sessions, providing one-on-one support to students in troubleshooting and optimizing their systems.
- Evaluated student projects and provided constructive feedback to enhance their technical and analytical skills.

Sanofi Bangladesh *Dhaka, Bangladesh*

Trainee Engineer *2019*

- Worked there as an intern as a part of our academic curriculum.
- Explored their plants, research and development wing, product design, and engineering section to gain insight into the industrial production of drugs.

Publications & Manuscripts

- Woods, D.P., Adams, G.M., **Mozumder, R.**, Dang, W., Chen, A.Y., Chevée, M., Constantinidis, C. and Gonzales, D.L., 2026. Repeatable, low-drift recordings in behaving non-human primates using flexible microelectrodes. Nature Communications (2026). In press.
- **Mozumder, R.**, Wang, Z., Dang, W., Zhu, J., Hammond, B.M., Machado, A. and Constantinidis, C., 2026. Asynchronous firing and off states in working memory maintenance. Cell Reports, 45(1).
- Karimi, A., **Mozumder, R.**, Schoenhaut, A.M., Rausis, O.G., Wallace, M.T., Ramachandran, R. and Constantinidis, C., 2025. Integration of audiovisual motion in dorsolateral prefrontal cortical neurons. Journal of neurophysiology, 134(4), pp.1097-1110.
- **Mozumder, R.**, Chung, S., Li, S. and Constantinidis, C., 2024. Contributions of narrow-and broad-spiking prefrontal and parietal neurons on working memory tasks. Frontiers in Systems Neuroscience, 18, p.1365622.
- **Mozumder, R.** and Constantinidis, C., 2023. Single-neuron and population measures of neuronal activity in working memory tasks. Journal of Neurophysiology, 130(3), pp.694-705.

Presentations

Jun 2026	**Optogenetics in non-human primates**, Session instructor	*MaDeLaNe Workshop*
Apr 2026	**Effects of activating Somatostatin interneurons in primate prefrontal cortex**, Talk	*VIRAL symposium*
Nov 2025	**Optotagging SST interneurons in Macaque Prefrontal Cortex Reveals Heterogeneous Physiology and Task Modulation**, Poster presentation	*Society for Neuroscience*
Sep 2025	**Optotagging SST interneurons in Macaque Prefrontal Cortex**, Poster presentation	*VBI Annual Retreat*

Skills

Programming/Scripting	Python, Matlab, R, C
Softwares/Packages	Matplotlib, Numpy, Scipy, Pandas
Deep Learning	Pytorch, TensorFlow
Medical Image Analysis	3D Slicer, AFNI, FSL
Data Acquisition	Open Ephys
Simulation & Design	Matlab & Simulink, Proteus, ANSYS, Solidworks, LabVIEW
Miscellaneous	Exploratory Data Analysis, Data Visualization, LaTeX, Microsoft Office, Google Workspace, Adobe Illustrator
Soft Skills	Time Management, Teamwork, Problem-solving, Documentation, Experiment Design, Presentation.

Academic Projects

2024	**Behavioral Decoding Using Recurrent Neural Networks (RNNs)**, We compared different types of RNNs to decode a behavioral feature from neural activity. We also explored how temporal history affects current decision-making, and contributions of different brain regions. Project presentation	*Neuromatch*
2023	**Non-linear Dimensionality Reduction and Visualization of Latent Space Representation of Neural Data in a Working Memory Task**, In this work, a transformer-based autoencoder model is proposed to perform dimensionality reduction of neural data. Visualization of the encoded latent space shows that the clusters of correct and error trials separately. These results indicate the potential of this kind of analysis over linear methods.. Project report	*VU*
2023	**Preprocessing and Higher-level Analysis on a fMRI Dataset to Visualize the Effect of Shape and Word Stimuli**, We investigated how the brain responds to word recognition and how task effects in the Visual Word Form Area (VWFA) differ from those in the other areas of the brain. Our main goal was to preprocess raw fMRI data, perform a GLM analysis to understand how the brain responded to various stimuli, and end with a functional connectivity study to understand the correlation of the brain regions with the VWFA. Project report	*VU*
2021	**Deep Learning Based Decoding of Motor Imagery for Practical Brain-Computer Interfaces**, We used transfer learning for Brain-Computer Interfaces (BCI), addressing motor imagery decoding. The challenge lies in transferring from multiple data sets which use different EEG setups comprising hundreds of users, to a set of new users that need to be up and running with only minutes' worth of training data. A novel architecture, ENET - Conv, was proposed to address the stated problems.	*BUET*
2018	**Developing a non-invasive blood glucose monitoring device**, An IR transmitter was used to transmit IR light through human finger and an IR receiver was used to receive the transmitted light after being absorbed by the blood glucose molecule. Then linear regression analysis was performed to measure the blood glucose level.	*BUET*

Honors & Awards

Aug 2026	**MedTech Design Sprint by Nissha Medical Technologies**, Runner-up	*Nashville, TN, USA*
Nov 2026	**Travel Grant**, VUSE	*Nashville, TN, USA*
2022-27	**Engineering Graduate Fellowship**, VUSE	*Nashville, TN, USA*
2020	**Design for Life Competition**, 4th Place	*Dhaka, Bangladesh*
2017,18,20	**Dean's List Honor**, awarded for CGPA>3.75 during an academic year	*Dhaka, Bangladesh*
2008-2020	**Board Scholarship by Intermediate and Secondary Education Board**, 4 time awardee	*Chittagong, Bangladesh*

Leadership & Voluntary Experiences

Mar 2026	**Brain Blast by Vanderbilt Brain Institute**, Volunteer	*Nashville, TN, USA*
'25, '26	**Multi-Area, high-Density, Laminar Neurophysiology (MaDeLaNe) Workshop**, Tutor	*Nashville, TN, USA*
2023-24	**Biomedical Engineering Graduate Student Association**, Treasurer	*Nashville, TN, USA*
2016-21	**Badhan, Ahsanullah Hall Unit, BUET Zone**, Active member	*Dhaka, Bangladesh*
2019	**Ahsanullah Hall Kali Puja Committee**, President	*Dhaka, Bangladesh*
2020	**Ahsanullah Hall Bani Archana Committee**, Secretary	*Dhaka, Bangladesh*

Standardized Tests

Oct 2021	**GRE**, Quant-166 (86th percentile), Verbal-152 (53rd percentile), AWA-4 (54th percentile)	*Dhaka, Bangladesh*
Sep 2021	**TOEFL**, Overall-105 (Reading-28, Listening-29, Speaking-23 , Writing-25)	*Dhaka, Bangladesh*

Languages

English: Professional fluency **Bengali:** Native Proficiency, **Hindi:** Professional fluency, **Spanish:** Basic Knowledge.

References

Christos Constantinidis, Ph.d.: Professor, VU, E-mail: christos.constantinidis.1@vanderbilt.edu
Ramnarayan Ramachandran, Ph.d.: Professor, VUMC, E-mail: ramnarayan.ramachandran@vumc.org
Muhammad Tarik Arafat, Ph.D.: Professor, BUET, E-mail: tarikarafat@bme.buet.ac.bd